**INDIAN BANK**

# Understanding Customer Behaviour Across Cards, Transactions, and Merchants

*Exploratory Data Analysis Report*

Dataset: Customer, Card, Transaction, and Merchant Records (100,000 transactions, 25,000 customers)

# 1. Business Context and Objective

The Indian banking and digital payments ecosystem has become increasingly diverse, with customers interacting with the bank through multiple card products, payment channels, merchant categories, and transaction types. As the customer base grows, understanding how different customers use banking products and how their transaction behaviour varies becomes increasingly important for improving customer engagement and identifying business opportunities.

This report analyses customer, card, transaction, and merchant data to uncover meaningful behavioural patterns and to identify which characteristics are associated with differences in transaction outcomes, spending patterns, and merchant interactions.

## Objective

To understand the different behavioural patterns within the bank's customer base and identify which customer, card, transaction, and merchant characteristics are associated with meaningful differences in customer behaviour and business outcomes — giving banking stakeholders a clearer understanding of their customers and highlighting areas where engagement, card usage, and overall experience could be improved.

## Primary Metrics Analysed

- Transaction Amount (spend behaviour) — total and average spend
- Transaction Status (success, decline, and failure rate)
- Fraud Flag (fraud rate) — risk and outcome quality
- RFM Segment — a composite behavioural metric (Recency, Frequency, Monetary) used to group customers by engagement level

## Data Sources

Four linked datasets were used: Customer_data (25,000 customers), Card_Data (32,457 cards), Transaction_Data (100,000 transactions), and Merchant_table (merchant profiles). Tables were validated for referential integrity — all Customer_ID, Card_ID, and Merchant_ID references in the transaction data matched their respective master tables, with no orphaned records found.

# 2. Hypotheses

Before conducting the analysis, eight hypotheses were framed to test whether specific customer, card, transaction, and merchant characteristics meaningfully influence behaviour and outcomes. Each is tested against the data in the sections that follow and summarised here.

| # | Hypothesis | Result |
|---|---|---|
| **H1** | Customer demographics (age, gender, income, occupation, location) influence transaction behaviour and RFM segment. | **Rejected** |

| # | Hypothesis | Result |
|---|---|---|
| **H2** | Digital payment method and channel usage (UPI, Net Banking, mobile) correlates with higher customer engagement. | **Confirmed** |
| **H3** | International transactions have different (lower) success rates than domestic transactions. | **Confirmed** |
| **H4** | Dormant customers show worse transaction outcomes (higher fraud/decline rate) than active customers. | **Confirmed** |
| **H5** | Card attributes (type, network, physical/virtual mode) affect transaction success rate. | **Rejected** |
| **H6** | Higher credit limits are associated with higher average spend per transaction. | **Confirmed** |
| **H7** | Merchant risk classification (High/Medium/Low) predicts fraud rate. | **Rejected** |
| **H8** | Fraud is more likely to be associated with failed/declined transactions than successful ones. | **Confirmed** |

# 3. Data Preparation and Methodology

## Data Quality Checks

- Verified referential integrity across all four tables — every Customer_ID, Card_ID, and Merchant_ID in the transaction table matched a valid record in its master table.
- Standardised date fields (Issue_Date, Expiry_Date, Customer_Since, Transaction_Date) into consistent formats for time-based analysis.
- Checked for duplicate Transaction_IDs and null values across key fields; no material data quality issues were found beyond formatting.

## RFM Segmentation Method

Customers were segmented using Recency, Frequency, and Monetary (RFM) analysis — a standard technique for grouping customers by behaviour rather than demographics:

- Recency: Days since each customer's last transaction, measured against the most recent date in the dataset (not the current calendar date, since the data itself is historical).
- Frequency: Total number of transactions per customer.
- Monetary: Total transaction value per customer.

Each dimension was scored 1-5 by quintile, then combined into four behavioural segments: Best Customers, Regular, At Risk, and Dormant, using score thresholds where 4-5 is treated as "high" and 1-2 as "low" engagement on each dimension.

### Rate vs. Headcount Correction

An important methodological check was applied throughout this analysis: raw customer counts can be misleading when segment sizes differ. For example, initial fraud-reason tables appeared to show "Best Customers" overrepresented in fraud cases — but this was a volume artefact, since Best Customers simply transact far more often. All fraud and outcome findings in this report are based on transaction-level rates (fraud transactions ÷ total transactions), not raw customer counts, to avoid this bias.

## 4. Customer Behaviour — RFM Segmentation

Customers were grouped into four behavioural segments based on how recently, how often, and how much they transact.

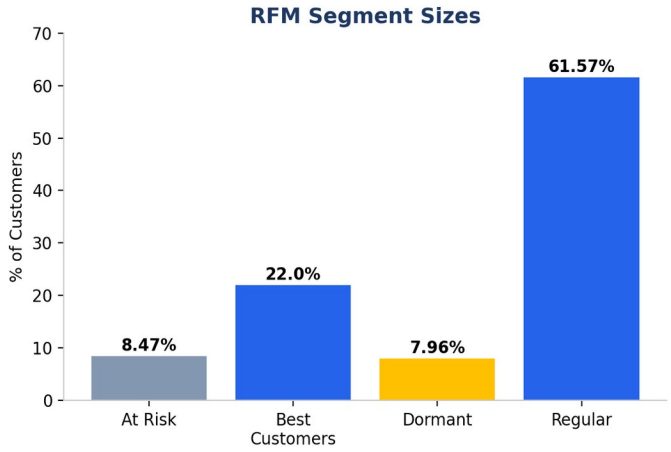



*Figure 1: RFM Segment Sizes — Regular customers form the majority (61.6%), followed by Best Customers (22.0%).*

### Does Customer Background Predict Behaviour?

Segment membership was tested against every available demographic attribute: gender, age, marital status, occupation, income tier, and state. In every case, the distribution stayed nearly identical — approximately 8-9% At Risk, 21-23% Best Customers, 7-8% Dormant, and 61-63% Regular, regardless of which group was examined.

> **Key Insight:** A customer's personal background (age, gender, income, occupation, location) does not explain their behavioural segment (H1: Rejected). Behaviour is driven by how customers transact, not who they are.

### Fraud and Decline Rate by Segment

While demographics showed no pattern, segment behaviour itself was strongly linked to transaction risk. Correcting for transaction volume (see Methodology), fraud rates by segment are:

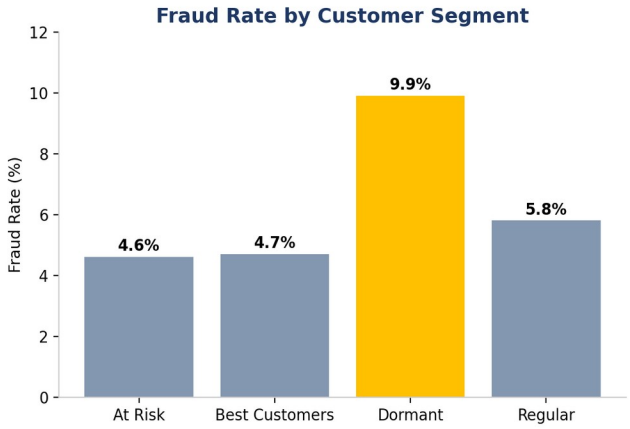



*Figure 2: Fraud Rate by Segment — Dormant customers show nearly double the fraud rate of every other group.*

**Key Insight:** Dormant customers show a fraud rate of 9.9%, roughly double every other segment (4.6-5.8%), and correspondingly higher decline rates (10.9% vs. 3.5-5.3%) and lower success rates (86.4% vs. 90.7-92.1%) (H4: Confirmed).

## 5. Why Are Dormant Customers at Risk? — Card Health

To understand the Dormant segment's elevated fraud and decline rate, card-level records were examined. A meaningful share of failed transactions for Dormant customers were tied to specific fraud/decline reasons: attempts on Blocked, Expired, or Lost cards — categories where Dormant customers were heavily overrepresented, while Best Customers were almost never affected.

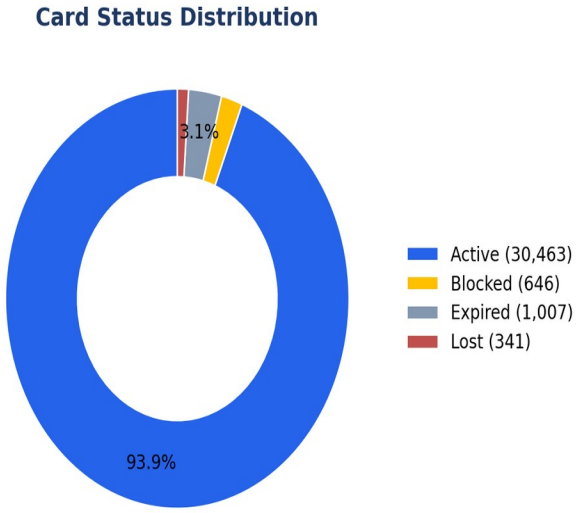



*Figure 3: Card Status Distribution — 94% of all cards are active; 6.1% are Blocked, Expired, or Lost.*

**Key Insight:** Across the full card base of 32,457 cards, 1,994 cards (6.1%) are Blocked, Expired, or Lost — a real, non-trivial pool of unusable cards. This suggests a meaningful share of Dormant customers may not have lost interest in the bank at all — they may

simply be unable to transact because their card needs renewal.

# 6. Transaction Outcomes

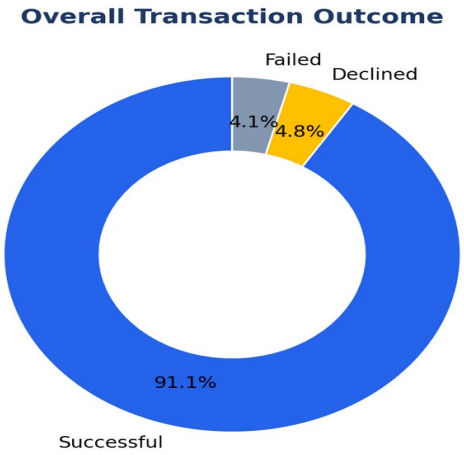



*Figure 4: Overall Transaction Outcome — 91.2% of all transactions succeed.*

## What Doesn't Affect Success Rate

Transaction success rate was tested against Transaction_Channel, Device_Type, Payment_Method, Card_Type, Card_Network, and Card_Mode (physical vs. virtual). In every case, the success rate stayed within a narrow band of approximately 89-92%, showing no meaningful pattern (H5: Rejected).

- Channel (ATM / Mobile App / Online Web / POS): 90.7% - 91.2% success — flat.
- Device Type (Android, iOS, Desktop, POS Terminal): 89.9% - 91.4% success — flat.
- Payment Method (Credit Card, Debit Card, Net Banking, UPI): 91.1% - 91.3% success — flat.
- Card Type, Card Network, Card Mode: all within 90.9% - 91.4% — flat.

## What Does Affect Success Rate — International Transactions

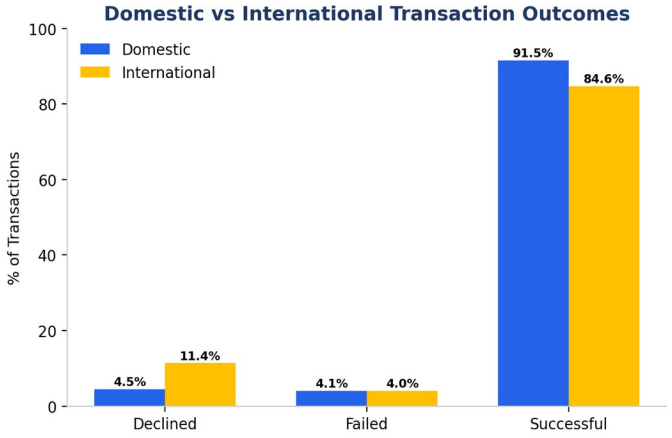


*Figure 5: Domestic vs. International Outcomes — International transactions fail/decline at nearly double the rate of domestic ones.*

**Key Insight:** Domestic transactions succeed 91.5% of the time; international transactions succeed only 84.6% of the time (H3: Confirmed). This points to added friction from stricter fraud checks, currency or network issues, or bank-side authorisation rules for foreign transactions.

# 7. Fraud Detection Effectiveness

Fraud was examined against Transaction_Status to understand whether fraudulent activity is being caught before completion.

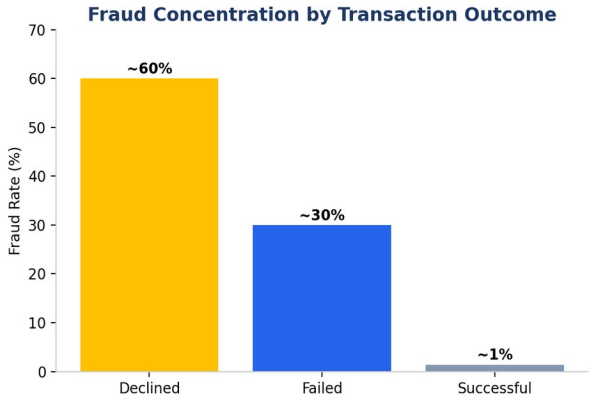



*Figure 6: Fraud Concentration by Outcome — fraud is heavily concentrated in transactions that fail or get declined.*

**Key Insight:** Approximately 60% of Declined transactions and 30% of Failed transactions are fraud-flagged, compared to only 1-2% of Successful transactions (H8: Confirmed). This indicates the bank's fraud controls are effectively intercepting most fraudulent activity before it completes.

## Merchant Risk Classification

Fraud rate was also tested against Merchant_Rating and Merchant_Risk_Level (High/Medium/Low). Neither showed a meaningful difference — fraud concentration by outcome (Declined ~59-63%, Successful ~1.1-1.7%) remained nearly constant across all rating and risk tiers.

**Key Insight:** Merchant_Risk_Level does not meaningfully predict fraud rate (H7: Rejected) — High, Medium, and Low risk merchants show almost identical fraud patterns, suggesting the bank's current merchant risk classification may need recalibration.

# 8. Credit Limit and Spending Behaviour

Average transaction value was examined across credit limit tiers to test whether a customer's credit limit genuinely drives how much they spend per transaction.

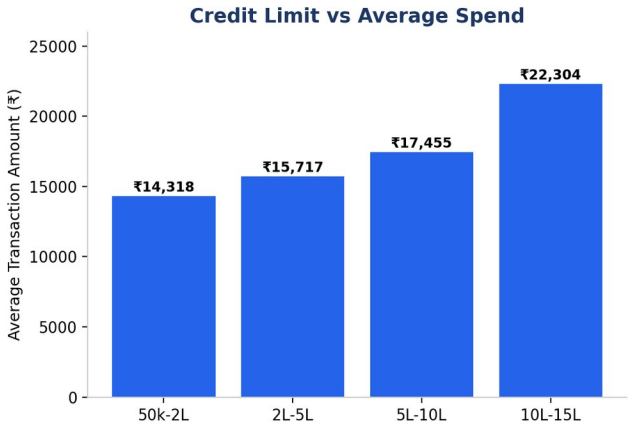



*Figure 7: Credit Limit vs. Average Spend — average transaction size rises steadily with credit limit.*

**Key Insight:** Average transaction value increases from ₹14,318 in the lowest credit limit tier (₹50k-2L) to ₹22,304 in the highest tier (₹10L-15L) — a 56% increase (H6: Confirmed). This is one of the few card-related attributes that meaningfully drives spend behaviour.

## 9. Merchant Category — Where the Money Goes

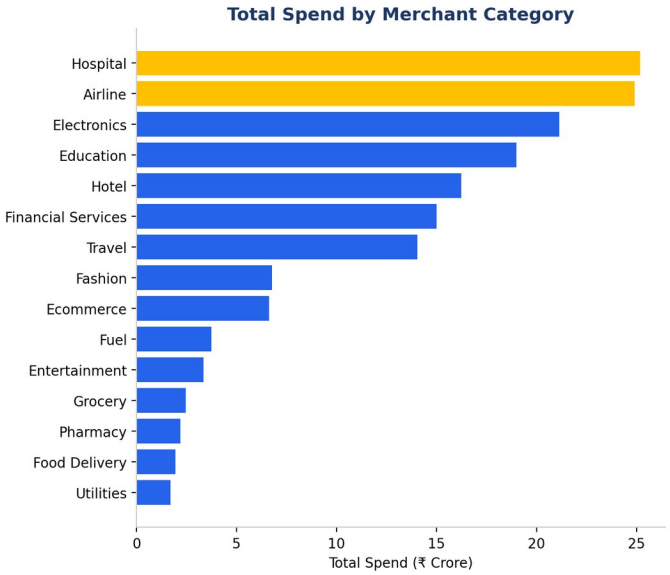



*Figure 8: Total Spend by Merchant Category — Hospital and Airline drive the highest total spend despite not having the highest transaction volume.*

Everyday categories such as Fuel, Hotel, and Travel are used frequently but in small amounts per transaction. Hospital, Airline, and Electronics are used less often, but each transaction is large — meaning these occasional, high-value categories drive a disproportionate share of total spend.

- Highest average transaction value: Electronics (₹37,333) and Hospital (₹35,469).
- Lowest average transaction value: Food Delivery (₹2,986) and Utilities (₹3,385).
- Highest transaction volume: Hotel (8,270 transactions) — but not the highest total spend.

# 10. Time-Based Patterns

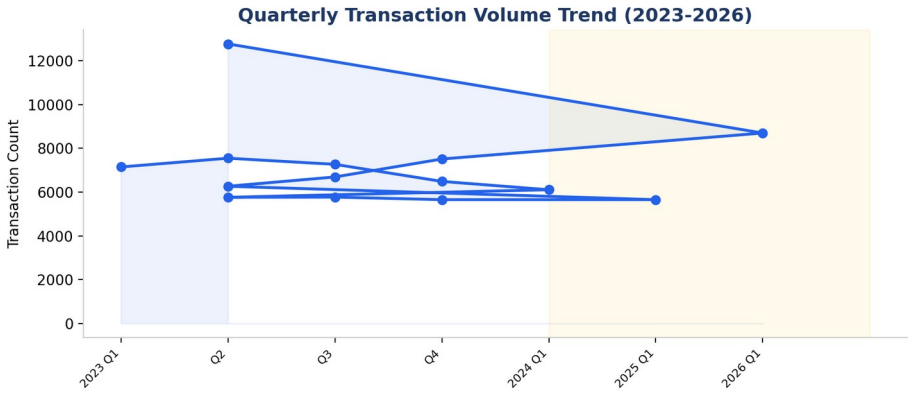



*Figure 9: Quarterly Transaction Volume (2023-2026) — a dip through 2024, steady recovery in 2025, and a sharp rise in 2026 Q2.*

Transaction volume dipped through 2024 relative to 2023, before recovering steadily across 2025 and into 2026, with a notable spike in 2026 Q2. Note: 2026 Q3 reflects a partial quarter, as the dataset ends mid-period, and should not be read as a decline.

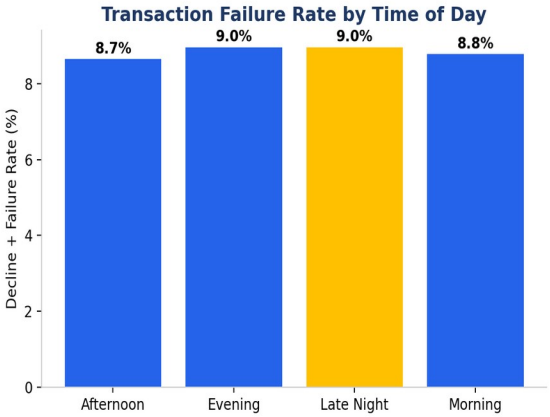



*Figure 10: Failure Rate by Time of Day — late-night transactions show a modestly elevated failure/decline rate.*

> **Key Insight:** Late-night transactions (12AM-6AM) show a combined decline/failure rate of approximately 10.0%, compared to 8.5% for afternoon transactions — consistent with known fraud patterns such as velocity and geographic anomalies that disproportionately occur outside standard hours.

# 11. Spending Patterns by Customer Tier and Age

Average transaction value was examined by customer tier (Basic/Gold/Platinum/Standard) crossed with age group, to determine which factor more strongly explains spending differences.

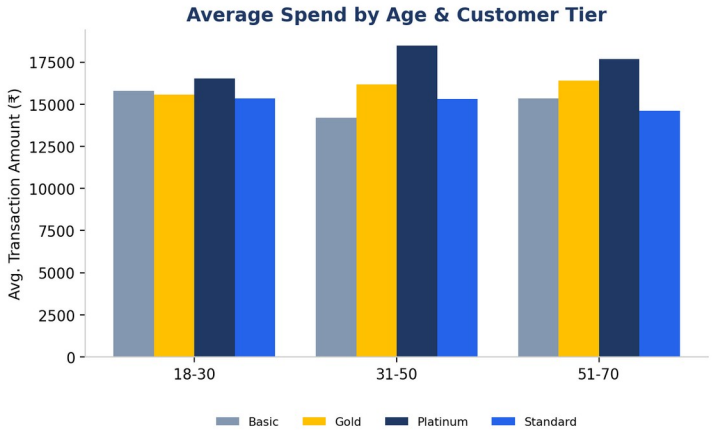



*Figure 11: Average Spend by Age and Customer Tier — Platinum customers consistently outspend other tiers within every age group.*

- Customer tier is a stronger differentiator of spend than age: Platinum customers average ₹17,980 per transaction vs. ₹15,047 for Standard, while age-group averages are far closer together (₹15,762 - ₹16,834).
- 31-50 Platinum customers spend ₹76,772 per customer overall — the highest of any age-tier combination — driven by transaction size, not frequency (4.15 transactions/customer, nearly identical to Standard's 4.17).
- Higher income does not always mean higher spend: 51-70 Platinum customers have the highest average income (₹38.3L) but a lower average transaction value than 31-50 Platinum customers (₹17,692 vs. ₹18,487).

## 12. Summary of Key Findings

- Customer demographics (age, gender, income, occupation, location) do not predict behavioural segment or engagement — how customers transact matters far more than who they are.
- Digitally engaged customers (UPI, Net Banking, mobile app users) are more likely to be highly engaged ("Best Customers") than card-present or POS users.
- Dormant customers carry roughly double the fraud and decline rate of every other segment, and a significant share of this traces to expired, blocked, or lost cards rather than genuine disengagement.
- International transactions fail/decline at nearly double the rate of domestic transactions.
- Fraud is heavily concentrated in Declined and Failed transactions (not Successful ones), indicating the bank's fraud controls are working effectively.
- Higher credit limits are associated with meaningfully larger transactions — a 56% increase in average spend from the lowest to highest limit tier.
- Card attributes (type, network, physical/virtual) and merchant risk classification show little to no relationship with transaction outcomes or fraud, suggesting these are not useful predictors in their current form.
- Hospital, Airline, and Electronics categories — while used less frequently — drive the largest share of total customer spend due to high per-transaction value.

# 13. Recommendations

### 1. Prioritise Card Renewal for Dormant Customers

Since a significant share of Dormant customers' failed transactions trace to expired, blocked, or lost cards, the bank should launch a targeted card-reissue and renewal campaign for this segment before assuming disengagement requires a loyalty or win-back offer. This is a low-cost, high-impact operational fix.

### 2. Encourage Digital Payment Adoption

UPI and Net Banking users show consistently higher engagement than card-present users. The bank should promote digital payment adoption, particularly among Regular and At Risk customers, to shift more of the customer base toward higher-engagement channels.

### 3. Review International Transaction Friction

Given the significant success-rate gap between domestic and international transactions, the bank should audit its cross-border authorisation rules to determine whether current fraud/risk checks are appropriately calibrated or are causing avoidable failures for legitimate customers.

### 4. Explore Credit Limit Increases for High-Spend Customers

Given the clear positive relationship between credit limit and transaction size, selectively increasing limits for high-spend, low-risk customers could be a natural lever for revenue growth.

### 5. Recalibrate Merchant Risk Classification

Since Merchant_Risk_Level does not currently predict fraud rate, the bank's risk-scoring methodology for merchants should be reviewed and refined to better reflect actual fraud exposure.

### 6. Investigate Late-Night and POS Channel Friction

The modestly elevated failure rate for late-night transactions and lower engagement seen among POS-channel customers both warrant closer operational review to identify and reduce avoidable friction.

# 14. Limitations

- Dormant is the smallest RFM segment by transaction volume (approximately 3% of transactions), so its fraud and decline rates are based on a comparatively thin sample and are more sensitive to individual cases than larger segments.
- Some age-tier combinations (e.g., 31-50 Basic, with only 14 customers and 56 transactions) have too few data points for fully reliable conclusions and should be treated as directional rather than definitive.
- The 2026 Q3 figures reflect a partial quarter, as the dataset appears to end mid-period, and should not be interpreted as a decline in activity.
- This analysis is based on historical transaction data and pattern correlation; it does not establish causation. Recommendations should be validated with further operational or controlled testing before implementation.

- The RFM segmentation thresholds (quintile-based, with scores 4-5 treated as "high" and 1-2 as "low") reflect a reasonable, standard approach but are one of several valid ways to define behavioural segments; results may shift slightly under different threshold choices.